09/01/18

TRANSPORTATION MODEL.

A model is ~~an transportation~~ abstraction of reality.
4 Principles of Production.

1. Quality product
2. Quantity
3. Available at the right time.
4. Right place.

Transportation model ~~is~~ the means that describes various methods of transporting goods from different source (origin where the product are been supplied) to Destination (where the product are been demanded or demand point).

For you to solve a transportation model, it has to be a balanced transportation model.

A balanced transportation model is made up of equal demand & supply i.e where what is going to be supplied at various origin is equal what is supplied at various destination otherwise it is unbalanced.

To resolve an unbalanced transportation model, you will create a dummy row or column to absolve the difference the demand & supply.

Dummy: in transportation model, a dummy is a row or column created with a zero cost of transportation to absorb the difference btw demand & supply or other to make a transportation model balanced.

Methods Of Solving A Transportation Model.

1) Initial basic feasible solution method which comprises of 3 means
   a) North west Corner method (2) least cost method (LCM)
   c) Vogel Approximation method (VAM).

3) Optimal Method: It is made up of 2 means.
   (1) Stepping stone algorithm method.
   (2) Modified distribution method.

ILLUSTRATION 1: A company has 3 production facility X_1, X_2, X_3 with production capacity 700, 900, 1800 tons of the output per week respectively. They intend to be shipped to 4 warehouses D_1, D_2, D_3, D_4 with requirements of 500, 800, 700, 1400 tons per week respectively. The transportation cost in naira per unit from factories to warehouse are given in the table below

	D_1	D_2	D_3	D_4	Supply
X_1	19	30	50	10	700
X_2	70	30	40	60	900
X_3	40	8	70	20	1800
Demand	500	800	700	1400	3400

You are required to:
1) formulate as a linear programming model.
(2) Solve for initial basic feasible soln
(3) Calculate the cost of initial basic feasible soln & compare.

Solution

	D_1	D_2	D_3	D_4	Supply
X_1	19	30	50	10	700
X_2	70	30	40	60	900
X_3	40	8	70	20	1800
Demand	500	800	700	1400	3400

4 steps in formulating a linear programming model.
1) Identify the decision variables
(2) State the obj function
(3) Applying the constraint
4) State the un negativity function

S = Unknown quantity to be distributed

$$\text{Minimize} \begin{cases} 19S_{X_1D_1} + 30S_{X_1D_2} + 50S_{X_1D_3} + 10S_{X_1D_4} + \\ 70S_{X_2D_1} + 30S_{X_2D_2} + 40S_{X_2D_3} + 60S_{X_2D_4} + \\ 40S_{X_3D_1} + 8S_{X_3D_2} + 70S_{X_3D_3} + 20S_{X_3D_4} \end{cases}$$

Supply Constraint

$$S_{X_1D_1} + S_{X_1D_2} + S_{X_1D_3} + S_{X_1D_4} = 700$$
$$S_{X_2D_1} + S_{X_2D_2} + S_{X_2D_3} + S_{X_2D_4} = 900$$
$$S_{X_3D_1} + S_{X_3D_2} + S_{X_3D_3} + S_{X_3D_4} = 1800$$

$S_{x1D_1} + S_{x1D_2} + S_{x1D_3} = 520$
$S_{x2D_2} + S_{x2D_3} + S_{x2D_4} = 800$ } Demand Constraint
$S_{x1D_3} + S_{x2D_3} + S_{x3D_3} = 700$
$S_{x3D_4} + S_{x3D_4} + S_{x3D_4} = 1400$

$S_{ij} \geq 0$ for all i, j (non negativity function).

MARKOV DECISION MODEL.

It involves creating system that changes over time i.e system that changes with time.

Many real problems can be modeled by using Markov chains. A markov chain is a probabilistic tools that is used for representing system that changes over time. In our discussion of Markov chain, we shall illustrate them use in analysing several practical problems by using an example i learning centres. When we have finished discussing this topic, you should be able to pull up.

i) Gain the definition of Markov chain
ii) Use & explain the terms used in Markov chain
iii) Draw and explain the transition diagram
iv) Derive transition matrix from transition diagram.
v) Explain what is meant by step transition.
vi) Compute the steady state probabilities
vii) Compute future market shares.
viii) Calculate the mean return time.

THE BRAND SWITCHING MODEL.

Mr. Jones the owner of a large Supermarket has been considering the brand switching behaviours of the customers in the purchase of a product, say Product A and B, which are alternative substitute to one another. The marketing staff of the supermarket has collected data over a period of time that gives the ffg trend in the pattern of the brand switching of the customers.

i) A customer that purchase Product A at the beginning of one month, he also purchases Product A at the beginning of the 2nd ffg month with a probability of 0.6 while he will switch over to Product B at the beginning of a ffg month with the probability of 0.4

ii) A customer that purchase Product B at the beginning of one month & purchase B at the beginning of the second month with a probability of 0.7, while the customer will switch over to Product A at the beginning of the ffg month with the probability of 0.3.

Mr. Jones would like to use the information collected from the market research data to generalize and analyze buyer behaviour pattern in the future.

Terms in Markov Chain:

- TIME: You are talking abt the specific time an activity is taking place. In the illustrations below, A was bought at the beginning of each month, it going to be horizontal. Activity is taking place at discrete intervals number only.



Difference btw Markov process & Markov chain

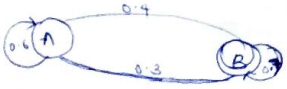
A markov chain is a discrete Markov process. It is a subset of Markov process. Process is taken place continuously over time.

STATE: It indicates a particular product that the customer is purchasing at a particular point in time. When a customer purchase product A in that customer is in state A same as B. A & B remain the state of the system

TRANSITION: It means movement of changing state

STATE TRANSITION: Moving from one state to another state which may also mean the state he was before.

TRANSITION DIAGRAM: Using a diagram to represent the movement of the product.



TRANSITION MATRIX:

P = This Month \ Next Month	A	B
A	0.6	0.4
B	0.3	0.7

It determines the behavioural pattern of the customer. P becomes the transition matrix

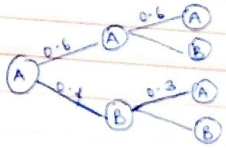
Ques

If a customer decides to buy A the first week, B the second & the third week, what is the probability that the customer will buy product A?

Ans: 0.3

Ques:

If a customer buys Product A the first week, what is the probability he buys that same Product in the 2nd month time

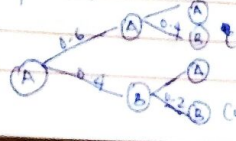


$(0.6)(0.6) = 0.36$

$+ = 0.48$

$(0.4)(0.3) = 0.12$

Same question, but it is B

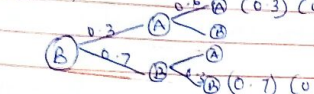


$(0.6)(0.4) = 0.24$

$+ = 0.52$

$(0.4)(0.7) = 0.28$

Suppose a store originally with B; what is the probability he buys product A in the 2nd time?

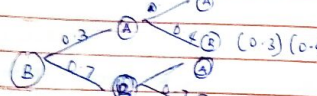


(0.3)(0.6) = 0.18

(0.7)(0.3) = 0.21

+ = 0.39

then from B to B is?



(0.3)(0.4) = 0.12

(0.7)(0.7) = 0.49

+ = 0.61

2 MONTHS TIME

P^2 = This month

	A	B
A	0.48	0.52
B	0.39	0.61

If you keep doing this for years it becomes cumbersome so we use Matrix multiplication operation

$P^2 = P \cdot P$

$$= \begin{matrix} \\ A \\ B \end{matrix} \begin{bmatrix} A & B \\ 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix} \times \begin{matrix} A \\ B \end{matrix} \begin{bmatrix} A & B \\ 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix} = \begin{matrix} \\ A \\ B \end{matrix} \begin{bmatrix} A & B \\ 0.36+0.12 & 0.24+0.28 \\ 0.18+0.21 & 0.12+0.49 \end{bmatrix}$$

$$P^2 = \begin{matrix} \\ A \\ B \end{matrix} \begin{bmatrix} A & B \\ 0.48 & 0.52 \\ 0.39 & 0.61 \end{bmatrix}$$

To find the probability in 3 month time

$P^3 = P \cdot P^2 \times$

$P^2 \cdot P$ ✓

$$\begin{matrix} \\ A \\ B \end{matrix} \begin{bmatrix} A & B \\ 0.48 & 0.52 \\ 0.39 & 0.61 \end{bmatrix} \times \begin{matrix} A \\ B \end{matrix} \begin{bmatrix} A & B \\ 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$$

$$\begin{matrix} \\ A \\ B \end{matrix} \begin{bmatrix} A & B \\ 0.288+0.156 & 0.192+0.364 \\ 0.234+0.183 & 0.156+0.427 \end{bmatrix}$$

$$P^3 = \begin{matrix} \\ A \\ B \end{matrix} \begin{bmatrix} A & B \\ 0.444 & 0.556 \\ 0.417 & 0.583 \end{bmatrix}$$

DEFINITION.

A Markov chain is a process in which the probability of the occurrence of an event in the future depends only on current events and not on previous events

MARKET SHARE ANALYSIS.

Suppose that the shares of the market currently held by Product A & B, initially is 45% to 55%, then we can use this information to deduce the shares of the market to be controlled by the product in the future



$[S_A \; S_B]_0$

What will be the shares of the market of for 2 product in 2 month.

$$[S_A \; S_B]_2 = [S_A \; S_B]_0 \cdot P^2$$

$$= \begin{bmatrix} 0.45 & 0.55 \end{bmatrix} \cdot \begin{bmatrix} 0.48 & 0.52 \\ 0.39 & 0.61 \end{bmatrix}$$

$$= \begin{matrix} \\ A \\ B \end{matrix} \begin{bmatrix} A & B \\ 0.2160+0.2145 & 0.2340+0.3355 \end{bmatrix} \begin{matrix} \\ B \\ 0.4305 \; 0.5695 \end{matrix}$$

$$= \begin{bmatrix} 0.4305 & 0.5695 \end{bmatrix}$$

43.05% 56.95%

For 3 month.

$[S_A \; S_B]_0 \times P^3$

$$\begin{bmatrix} 0.45 & 0.55 \end{bmatrix} \times \begin{matrix} \\ A \\ B \end{matrix} \begin{bmatrix} A & B \\ 0.444 & 0.556 \\ 0.417 & 0.583 \end{bmatrix}$$

- etc.

INVENTORY ANALYSIS.

Inventory analysis

Suppose we have 5000 customers that virtually patronize the super market in the purchase of the 2 products every month with each customer buying a packet of product of either product A or B_0. Then what should be the inventory of the 2 products for the 3rd month

Soln.

for Product A

$S_{n_2} \times 5000$

Inventory for A = 0.4305×5000

= 2152.5

$S_{B_2} \times 5000$

Inventory for B = 0.5695×5000

= 2847.5

STEADY STATE.

We are talking in that point in time, irrespective of if a person buy product A or B in the long run.

Limit P^N

$N \to \infty = [\pi_A \quad \pi_B]$ with $\pi_A + \pi_B = 1$

$[\pi_A \quad \pi_B] = [\pi_A \quad \pi_B] P$

Where P is the initial transition matrix.

$[\pi_A \quad \pi_B] \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$

$= [0.6\pi + 0.3\pi_B \quad 0.4\pi_A + 0.7\pi_B]$

$\pi_A = 0.6\pi_A + 0.3\pi_B$ — (i)

$\pi_B = 0.4\pi_A + 0.7\pi_B$ — (ii)

Recall: $\pi_B + \pi_B = 1 \Rightarrow \pi_A = 1 - \pi_B$

In eqn (i)

$1 - \pi_B = 0.6(1 - \pi_B) + 0.3\pi_B$

$1 - \pi_B = 0.6 - 0.6\pi_B + 0.3\pi_B$

~~$1 - \pi_B = 0.6 - 0.3\pi_B$~~

~~$1 - 0.6 = 0.3\pi_B$~~

~~$0.4 = 0.3\pi_B$~~

$1 - 0.6 = \pi_B + 0.3\pi_B - 0.6\pi_B$

$0.4 = 1.3\pi_B - 0.6\pi_B$

$0.4 = 0.7\pi_B$

$\therefore \pi_B = 4/7$

Since $\pi_A = 1 - \pi_B$

$= 1 - 4/7 = 3/7$

	A	B
A	3/7	4/7
B	3/7	4/7

[There will be a Stable Probability]

MEAN RETURN TIME.

If some one buys product A, how long will it take him to buy that product A. So from the problems, it was said it at the beginning of each month.

Mean Return Time for A =

$\frac{1}{\pi_A} = \frac{1}{3/7} = \frac{7}{3}$ months.

Mean Return for B = $\frac{1}{\pi_B} = \frac{1}{4/7} = 7/4$ months

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REPLACEMENT ANALYSIS

Organizations exists in social technical context that are made up human and non human items. However, because nothing last forever, there is a need to replace or maintain the items at some point in time. Replacement simply means substituting for or putting back while replacement analysis is the process by which the various cost consequences involved in replacement are studied so as to identify or make optimal replacement decisions. In other to do this, there is a need to know the categorizations of items that exist in organizations.

CATEGORIES OF ITEMS IN ORG

Basically, there are 3 categories of items in organisations namely.

1) Sudden Failure items eg bulbs, fan belt, blocks, adaptors etc.

23/04/18

There are decision making environment

1) Decision making under certainty
2) ✓ ✓ Under Stochastic/Risk
3) ✓ ✓ ✓ Uncertainty

In every decision making, there are decision variable. Decision variable are factors that influence decision making & there are 2 types

i) Controllable decision variable
ii) Uncontrolled ✓ ✓

Controllable decision variables: are variables that can be controlled by the decision model.

2) Items that deteriorate: These items deteriorate with time or usage but the loss of efficiency can be eliminated if these are maintained or repaired eg car engine

3) Items that deteriorate/depreciate or efficiency until it fails eg tyres

ASSUMPTIONS OF REPLACEMENT ANALYSIS

1) To minimize cost
2) Constant replacement cost; The purchase price of the item remain the same through the period
3) Constant technological level.
4) Constant maintenance cost
5) Constant output quality
6) Performance level

COST ASSOCIATED WITH REPLACEMENT ANALYSIS

A) Sudden failure items.

1) Replacement cost; this is the purchase price of the item.
2) Consequential cost of failure: cost you incur if the items fail suddenly.
3) Labour cost; cost of replacement; it is the sum total of all cost involved in replacing the items

COST ASSOCIATED WITH DETERIORATING ITEMS.

1) Maintenance cost
2) Depreciation cost

REPLACEMENT POLICIES

A) Sudden failure items; There are 2 types for replacement policies for sudden failure items:

1) Individual replacement policy
2) Group replacement policy.

1) Individual replacement policy; replacement the items as they fail one by one.
2) Group replacement policy.

2 types

1) Periodic group replacement policy
2) Group replacement periodically.

MECHANICS FOR INDIVIDUAL REPLACEMENT POLICY:

steps:

1. find the probability of failure per period. P_i

Find the probability of failure per period indicates as P_i. If the prob value as <1: <1 = Cummmulative probability. >1 = Cummmulative probability.

2) Find the average life of the item

Average life = $AL = \sum P_i t_i$

P_i - Probability for Period i

t_i - Period i

3) Find the number of items to be replaced

$N_r = \frac{N_o}{AL}$ - N_o = Initial no of item; AL = average life; initial no of items

4) Find the cost of individual replacement: $T_c = N_r(i)$

where:

P = Replacement cost = Purchase price

i_c = P + Consequential cost of items

MECHANICS OF GROUP REPLACEMENT.

We will use a tabulation approach.

steps:

1) Tabulate the time period.
2) Find the probability of failure per Period
3) Find the number to be replaced per period using the iffg formulae:

no of items to be replaced on 3rd period

$N_1 = N_o P_1$

$N_2 = N_o P_2 + N_1 P_1$

$N_3 = N_o P_3 + N_1 P_2 + N_2 P_1$

$N_4 = N_o P_4 + N_1 P_3 + N_2 P_2 + N_3 P_1$

$N_5 = N_o P_5 + N_1 P_4 + N_2 P_3 + N_3 P_2 + N_4 P_1$

P = Replacement cost/Purchase price.

4) Find the cummulative numbers to be replaced per period.
5) Find the cummulative individual related cost per period: This is equal to the cummulative number to be replaced per period x individual replacement cost per unit

6) Find FC = Initial no. of items x Group replacement cost.

7) Find the total cummulative replacement cost = Cummulative individual replacement cost per period + Fixed cost per period.

8) Find the average total cummulative replacement cost per period = Total cummulative replacement cost / respective period.

9) Select the least total average total Cummulative replacement cost and conclude that Group replacement should be done at the end of that period

10) NB: You go through this step if you are asked to Compare & recommend

Compare the cost of individual replacement policy with the cost of Group replacement policy & recommend the least cost policy as optimum.

Example; Assuming you are the maintenance officer for the faculty of Mgt Sience and you have been asked to recommend the appropriate replacement policy to be adopted to replace the florescent light bulbs in the facully classrooms blocks comprising of; Rm 1&2, 3, 4&5, 6, 7, 8&9

Assuming futher that the initial number of florescent light bulbs in all this classrooms is 200 and the cost of replacing them individually as they fail is ₦350 while the cost of replacing them as a group ₦10,000. You are required to analyze the situation completely given the ffg failure info per month.

Month	Prob of failure.
1	0.03
2	0.30
3	0.55
4	0.90
5	1.00
	2.78

It is cummulative probability by period as we move to

Months	Prob of failure	P_i	$P_i t_i$
1	0.03	0.03	0.03
2	0.30	0.27	0.54
3	0.55	0.25	0.75
4	0.9	0.35	1.40
5	1.00	0.1	0.50
	2.78	1.00	$\Sigma P_i t_i$ = 3.22

1) Find the probability of failure per period P_i per period.

2) $AL = \Sigma P_i t_i = 3.22$

3) Find no. of items to be replaced

$N_r = \frac{N_0}{AL}$

where $N_0 = 200$ $\quad \frac{200}{3.22} = 62.11 \approx 62.$

4) $I_c = N_r$ (62 (₦350) Individual replacement cost per unit.) = ₦217,000

25/1/18 MAINTENANCE OF DETERIORATING ITEMS

When dealing with deteriorating items, there are two different that can be used to determine the optimal point in time to replace such item. Two of such items are:

1) The minimum average total cost method
2) Differential Calculus method.

Example: The faculty of mgt sciences has inverter sets in it classroom blocks. The purchase price of the inverter set is ₦10,m, The resale value & maintenance cost over a 5yr period is given in data below:

Yr	Resale value (₦'000)	Maint cost (₦'000)
1	8000	300
2	6700	500
3	5000	800
4	2400	1400
5	1200	3000

Required: Find the optimal time that the faculty of mgt sciences should replace the inverter set without losing out on maximum cost of capital

HINTS: Use the average total cost method to solve a problem. An alternative way of determining when to replace deteriorating items is by using the differential Calculus. This entails differentiating the average total cummulating maintenance cost (Y/unit) with respect to the life span of the item(x) and then apply the turning points rule to solve the differentiated eqn with respect to x.

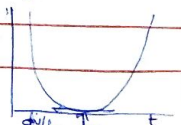
eg' Assuming the annual average total cummulative cost (Y) of the inverter sets in the faculty of mgt sciences expressed as a functions of their life span is given by the eqn:

$Y = 5x^2 - 60x + 2000$

Required: Determine the optimal replacement age for the inverter set using the differential Calculus method.

Soln

$\frac{dy}{dx} = nCx^{n-1}$



0 = turning point
dy/dx = 0

$\frac{dy}{dx} = 10x - 60$

at turning point $\frac{dy}{dx} = 0$

$10x - 60 = 0$

$10x = 60$

$x = 6$ yrs.

ASSEMBLY LINE BALANCE.

Org use various production systems to ensure that goods & services that meet customers needs are produced. In order to do this, they can apply either the flow process system or the job process system. The flow process is a production system in which materials flow is uninterrupted from the point of entry into the system to the point of exit while a job system is a production process/system in which goods are produced based on customers specification irrespective of the production process adopted sometimes in balances, in capacity and results utilization are experienced. Hence, there is a need to embark on ASSEMBLY LINE BALANCING.

Assembly line Balancing seeks to distribute task to workstation so as to optimize line operation. An assembly line is an arrangement of organizational resources whereby a product is assembled through the performance of specific successive operation on an incomplete unit as it passes by in a series of stages organized in a direct line. These stages are called Workstation. Assembly lines are mostly use to assemble complex products like bicycles, electronics, household appliances etc through mass assembly line methods are developed mainly for manufacturing processes, they can also be applied in other areas of biz that deals with project mgt. The decision problem of assembly line balancing entails

designing a appropriate work allocation to the work stations so as to minimize idleness and maximize efficiency. The goal of assembly line balancing is to find a way of grouping tasks in such a manner that all work stations

Have Balancing Soln Techniques

Though we have several line balancing techniques but can be categorized into 2:

Mathematical models which are precise exact approaches that gives optimal solutions

2) Heuristic methods are models based on rules of thumbs. The most popular one is the Kilbridge methods.

Operations research of terms

1) Cycle time: Time one tasks in a line. They are the smallest work units in which the total job content can be divided denoted by the symbol 'C'.

2) Cycle time (Cc)

$C = \frac{T - \text{Time given to complete all jobs}}{N_a - \text{No. of units in a production run}}$

T = the max time available for a task or group of tasks is be completed at each work station.

3) t_i is the time taken to complete task 'i'.

4) S_j is the total time taken by each work station

5) $S_j = C$ is a perfectly balanced line

6) $S_j \leq C$ is actual balance line situation

7) $S_c \geq \frac{\sum t_i}{N_a \times C}$

S_c

System Efficiency Measures.

This tells us the level of the system. It is obtained using the ffg:

$S_c = \frac{\sum t_i}{N_a \times C} \times 100$

Mechanics of Line Balancing

Since no cost type, we have fairly balanced assembly line situation, the Westerberg Kilbridge method will be adopted.

Below is the modified algorithm developed by DR MRS OBCECHI

STEP 1: Define all element task involved in getting a product.

STEP 2: State the precedence relationship (activity on arrow network or node)

STEP 3: Draw the network diagram.

STEP 4: Calculate the allowable or required work station cycle time using the formula: $C = T/H_r$

STEP 5: Compute the theoretical minimal no. of work stations needed using the formula $N = \frac{\sum t_i}{C}$

STEP 6: Construct no of ffg task table.

STEP 7: Construct assembly line balancing assignment or allocation table using ffg heading

Is Balance Delay: refers to idleness and it is a measure of system inefficiency.

STEP 3

Apply assignment rule as follows:

Primary rule: Assign the task with most number of followers.

b) Secondary rule: In case of a tie, assign the task with the longest operation time.

STEP 4: Repeat step 3 until all task have been assigned.

STEP 10: Evaluate system efficiency using the ffg formula

$$SE = \frac{\sum t_i}{N_a \times C} \times 100$$

STEP 11. Rebalance if necessary by going back to step 1

STEP 12: Stop

Example: Alhidam BMD has an assembly plant where rechargeable lamp fan are manufactured on a daily basis. Assume that her assembly plants operates on 7 hours daily. Assuming she produces 630 lamps daily. Required: find the balance time, the number of work station

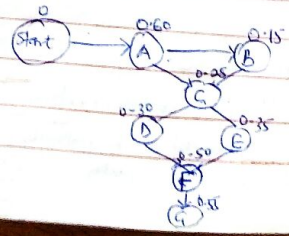
b) Compute balance delay given the ffg table

Task	Description	Precedence relationship	time
A	Depth component		0.60
B	Join component	A	0.15
C	Insert fan blades	A, B	0.25
D	Insert head light	C	0.30
E	Connect all wires	C	0.35
F	Fix complete all print	D, E	0.50
G	Table	F	0.55

$\sum t_i = 2.70$

STEP 2.

STEP 3



STEP 4: $C = \frac{T}{Nt}$

$T = 7 \times 60_{min} = 420$

$\therefore C = \frac{420}{630} = 0.67$ min

STEP 5: $N = \frac{\sum t_i}{C}$

$C = 0.67$ m

$\sum t_i = 2.70$

$2.70 / 0.67 = 4.03$

≈ 5 wks

STEP 6.

Task	Time	# of followers
A	0.60	6
B	0.15	5
C	0.25	4
D	0.30	2
E	0.35	2
F	0.50	1
G	0.55	0

STEP 7:

Primary rule:

workstation	Task	Task time	# of task	idle time	feasible remaining task	Task
1	A	0.60	6	0.07 hr		
2	B, C	0.15, 0.25	5, 4	0.33 hr	C, D, E, F	
3	D	0.30	2	0.27 hr, 0.02 hr	D, E	C
4	E, F	0.35, 0.50	2, 1	0.17 hr, 0.35 hr	F	D, E
5	G	0.55	0	0.12 hr, 0.67 hr		F

STEP 8:

$$S_E = \frac{\Sigma t_i}{V_A \times C} \times 100$$

$$\frac{2.70}{5 \times 0.67} \times 100$$

$$= 80.60\%$$

6.1.18 SEQUENCING & RUNTIME

SEQUENCING:

Introduction: Every organisation will like to utilize its productive system effectively & efficiently. So as to maximize profit by meeting delivery deadlines. Job involving many operations have to be performed & where as, there are limited resources in terms of plant & machines on which this job have been performed.

DEFINITION:

Sequencing is the selecting of appropriate order for a series of jobs to be done on a finite number of service facilities.

Assumption:

1) All machines can process only one job at a time.
2) Processing time is known & fixed.
3) When a job starts, it must be completed.
4) Preceeding operations must be completed before the succeeding one.
5) All jobs are in the specified order as soon as possible.
6) All jobs are ready for processing.
7) All of machines have different capacity & capability.
8) Processing time of a job have no relationship with the order in which the job is processed.
9) No time is wasted in shifting a job from one machine to another i.e the wasted time is 0.

TYPES OF SEQUENCING Problem

TYPE I - n jobs on one machine.
TYPE II - n jobs on two machines
TYPE III - n jobs on m machines

FREQUENCY TERMINOLOGIES

1) Machines: This is the facility which have some processing capabilities
2) Scheduling: This is when time & sequencing or processing a job is specified
3) Total Elapsed Time: It is the time that lapsed btw the starting of the first job & the finishing of the last job.
4) Idle time: This is the time the facility or machines are not used during the total elapsed time.
5) No Passing Rule: It refers to the rule of maintaining the order in which jobs are to be processed on a given machines i.e a particular job must first be complemented before any other one can start.

Example: The processing time for jobs A, B, C, D, E and the due date for this job on a machine shop are given in the table below

Job	Processing Time	Due date/day
A	9	16
B	7	20
C	5	25
D	8	20
E	6	40

These jobs are to be sequenced according to:

1) FCFS (First come first serve)
2) SPT - Shortest processing time
2) LS - least slack
4) EDD - Expected due date

Sol^n

FCFS - ① Repeat the first 2 column

Jobs	Processing
A	9
B	7
C	5
D	8
E	6

step 2: Cummulate process time & add them together. This is called the flow time

Job	Process Time	Flow time
A	9	9
B	7	16
C	5	21
D	8	29
E	6	35
	110	

Step 3: Depict due date

Job	Process Time	Flow time	Due Date
A	9	9	16
B	7	16	20
C	5	21	25
D	8	29	20
E	6	35	40
	110		

Comparison

Step 4: Do some comparing

Job A = No lateness

Job B = "

Job C = "

Job D = lateness

" E = No lateness.

Step 4: Comparison

Do some comparing btw the flow time with due date by checking for the difference.

Job	Process Time	Flow Time	Due Date	lateness
A	9	9	16	0
B	7	16	20	0
C	5	21	25	0
D	8	29	20	9
E	6	35	40	0
	35	110		9

Using FCFS

1) Total completion time will be 35 days

2) Average completion time = T. Flow time / no of jobs

T.Ft = 110 / 5 = 22 days

3) Average no. of work of the center = T. Flow / T. Process time = 110 / 35 = 3.14 jobs

4) Average job lateness = T. lateness / no of jobs

9/5 = 1.8 days

MODELS TO ANALYZE

LINEAR PROGRAMMING

A Scenario we find ourselves which is called Decision Making situation & situation:

1) Decision making under Certainty

2) " "

3)

In every decision made, there is always Decision Variable → they are criteria that affect your decision. They affect your final decision.

They are of 2

i) Controllable decision variable: they are under the control of the decision maker. They are also called options or alternatives.

ii) Uncontrollable decision variable: they are not under the control of decision maker but affect the final decision. The uncontrollable decision variable determine the environment you find yourself.

CERTAINTY SITUATION: If you have all information abt the Uncontrollable decision

BACK

If you don't have all the info abt the occurrence of the state of the uncontrollable decision it is called.

You don't have any info abt the occurrence of the state of the uncontrollable decision but you can apply probability distribution you have move from Uncertainty to Stochastic decision.

Uncertainty situation is in the biz world, but you don't have any info abt the occurrence of the state of the uncontrollable decision and you cannot apply probability distribution. So meaning you have to move from Uncertainty situation to Certainty situation.

Linear programming - is an OR objective model that uses a linear mathematical model to describe the problem

Linear programming is a mathematical model to allocate scarce or limited resources to an activity in an optimal manner, subject to the presence of restrictions.

Linearity: the highest power of the variable is 1

Proportionality: If you increase a input by 1 same unit output

Programmability: Flexibility: The variables can be added or removed

Certainty: all info you have is known

Deterministic: all variable are determined

Divisibility: variable can be fractional

Non-negativity: values are not allowed to have negative value. Zero is a value

Terminology

Assignment 11 Identify areas on forage where linear programming can be applied.

Graphical method - used to solve problems that involve only two variables

Linear programming - is an OR objective model that uses a linear mathematical model to allocate scarce or limited resources to an activity in an optimal manner

Concepts

1) Decision Variable: It is denoted for the activity the decision making can act upon. It is called a set of unknown quantities rep in the LP model by mathematical symbol $x_j, j = 1 \cdots \infty$

Z - maximization C - minimization

Two have to do with objectives

The next thing we look at x_j

2) Contribution: If it is maximization recall C - cost, for maximization C - contribution.

3) Resources: R_i. $i = 1 \text{ —— } \infty$

4) Constraints: b_i - limit of availability

5) Technological coefficient: a_{ij}

6) Basic variables 8) Surplus variable

7) Un basic variables 9) Slack variable

j - Column

i - row

In all other cases, where we use linear programming

* Any variable known is called deterministic model
* Any variable not known is called stochastic models. They involve probability

In LP, first carry out the activity before arriving at the analyses or decision.

Mathematical Formation of LP model

1) Determine or identify decision variable
2) Determine the objective function
3) Derive or determine the certainty eqn.

* The number of constraint eqn you will derive is dependent on the number of resources you will have
* In the constraint we have magnitudes

Maximization $\leq$

Minimization $\geq$

Equality $=$

If we have either $<, >, =$ it is called linear programming

Assymetric linear programming then the signs might be combined like $\leq \geq \leq$ or $= \leq =$

Constraint Constraint:

Sign	Interpretation	Example
$\geq$	greater than or equal to, at least, minimum number of, no less than	at least 25% of the total number houses built must be of 2 bedroomed
$>$	when you see words like greater than or more than that the greater than sign	more than 50% of construction workers must consist of first time buyers
$=$	when you see a word equal = this is applicable to constraint eqn	a) production balancing for a component what we make & what you buy of a certain must equal what you sell & what you store
$<$	less than or equal to, at most, maximum, number of no more than, use less or equal to sign	staff cannot work more than 50 hours per week

≤ | less than. | stiff must work less than 45hrs per week

f) Non negativity condition: A company engages in producing 2 products from 2 resources which are material & labour hour. Each units of product one will require one pund of material & 6h of labour. Each units of product 2 require 2 punds of materials & 6 materials of labour. The contribution to profit per unit of product 1 is ₹4 while each unit of product 2 contribute ₹5 to profit. The total material available for the production of this products is limited to 10 pounds while in all 36 labour hrs is available. Further sales product forcast has limited the product 1-4. Formulate the appropriate LP problem model for the given problem.

Soln

(1) Let x_j be the qty product 1 & 2 produced. where j - Product 1, 2, x_1, x_2

(2) $Z = ₹4x_1 + 5x_2$

Subject to (S.t)

R_1 - Material: $x_1 + 2x_2 \leq 10$

$6x_1 + 6x_2 \leq 36$

$x_1 \leq 4$

Non-negativity: $x_1, x_2 \geq 0$ or $x_1 \geq 0$, $x_2 \geq 0$

LEAST SLACK

STEP 1: Diff the due date & process time give us the slack.

Job	Due date	Processing time	Due date - Processing time = Slack
A	16	9	7
B	20	7	13
C	25	5	20
D	20	8	12
E	40	6	34

Step 2: Arrange the job, processing time and due date on the ascending order of the slack

Job	Processing time	Due date	Slack
A	9	16	7
B	8	20	12
D	7	20	13
C	5	25	20
E	6	40	34

STEP 3: Delete the slack, cummulate the processing time and put the value in between processing time & due date and tag it flow time and add up the flow time

Job	Process time	Flow time	Due date
A	9	9	16
D	8	17	20
B	7	24	20 (4)
C	5	29	25 (4)
E	6	35	40
	35	114	

STEP 4: Determine the late jobs by checking the due date with flow time using L.S

1) Total completion time = 35 days.

2) Average = Total flow time / no. of jobs = 114/5 = 22.8 days

3) Average Nos of work at the centre = Total flow time / Total process time = 114/35 = 3.26 days.

4) Average Job lateness: Total lateness / no of job = 8/5 = 1.6 days

5) Late Jobs (Tardy): 2 jobs. B & C.

BACK

EXPECTED DUE DATE (EDD)

STEP I: Arrange due date from least to highest with their corresponding jobs in that order

Job	Due date
A	16
B	20
C	20
D	25
E	40

STEP 2: Put their corresponding processing time in the middle of Table

Job	Process time	Due date
A	9	16
B	7	20
D	8	20
C	5	25
E	6	40

STEP III: Cummulate the process time and put the values between the process time & due date. Then add up the flow time as follows:

STEP IV: Determine the late jobs by checking the due date from flow time.

Job	Process time	Flow time	Due date	Late
A	9	9	16	0
B	7	16	20	0
D	8	24	20	4
C	5	29	25	4
E	6	35	40	0

Using EDD

① Total completion time = 35 days

② Average Completion time = $113/5 = 22.6$

③ Average Job lateness = $8/5 = 1.6$ days

④ Average No works in the centre = $\frac{113}{35} = 3.23$ jobs

⑤ Tardy Jobs = 2, D & C